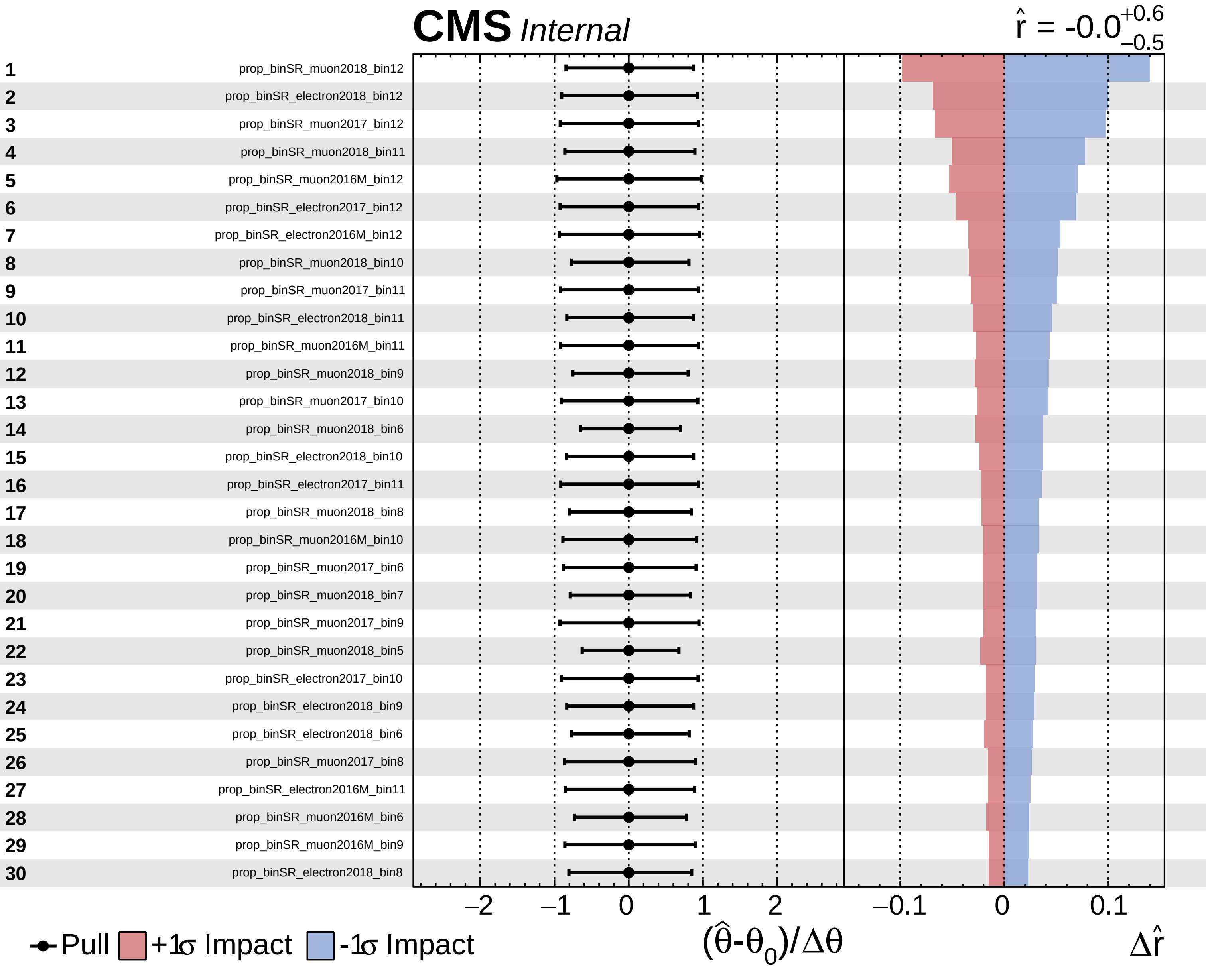
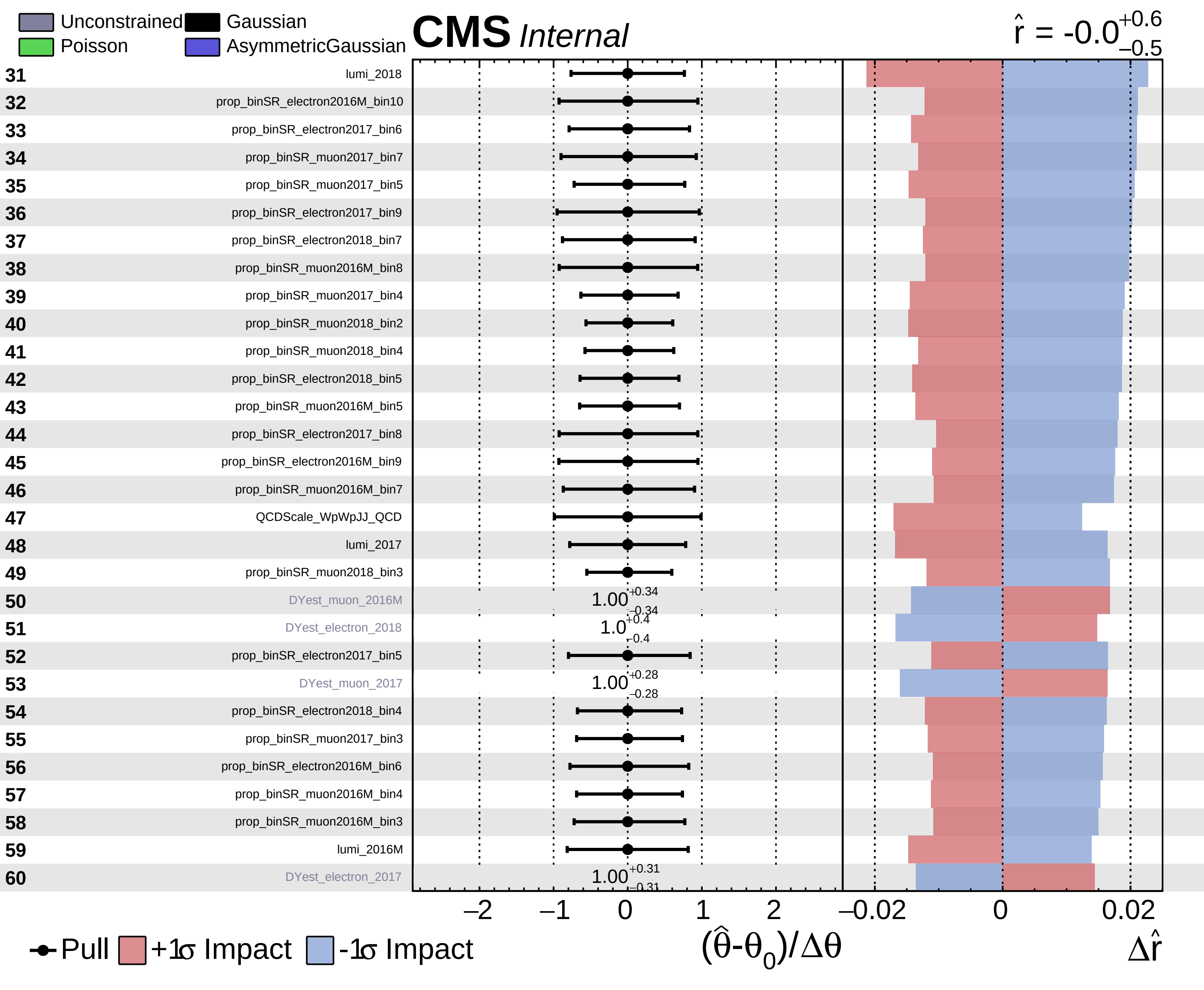
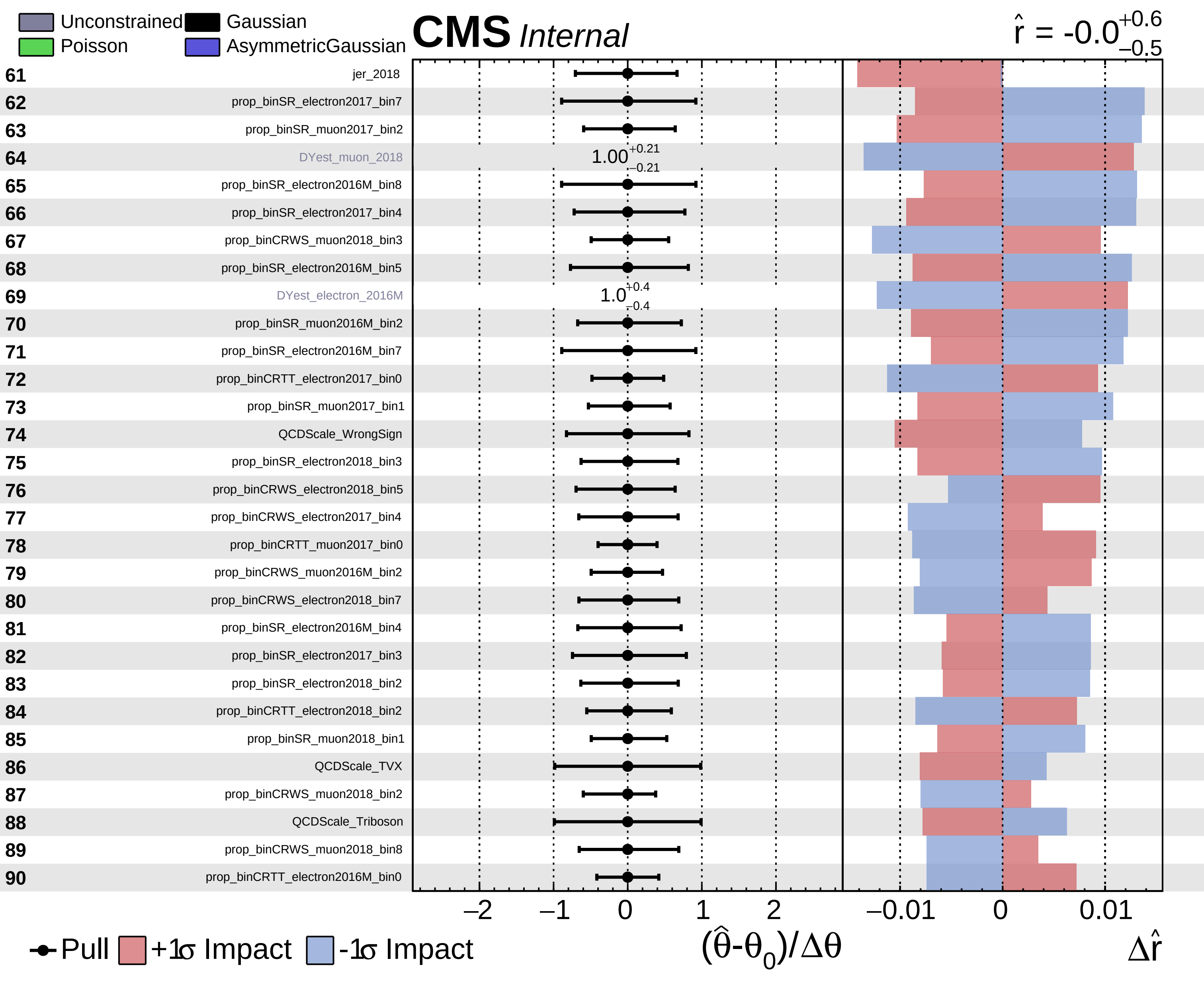
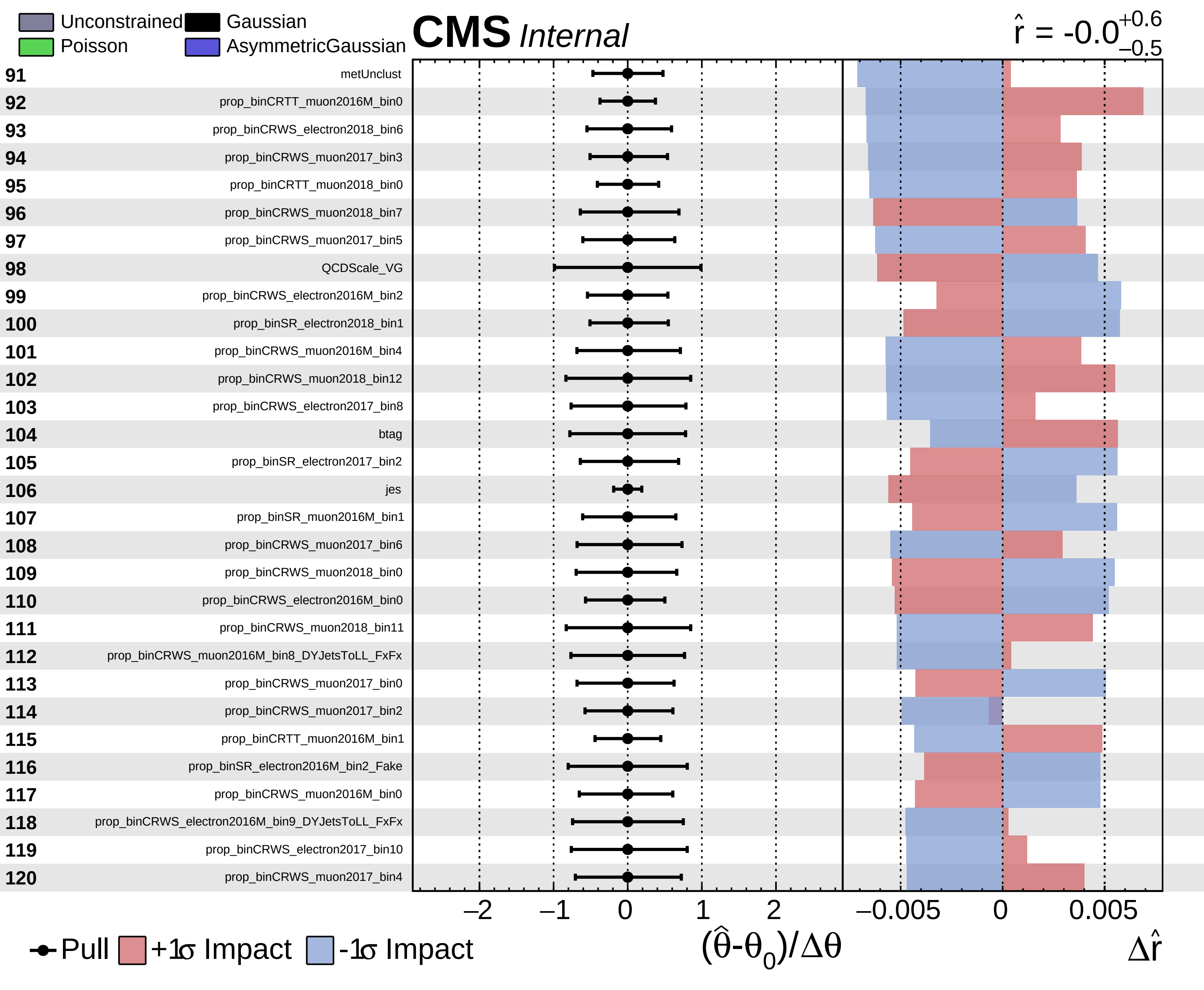


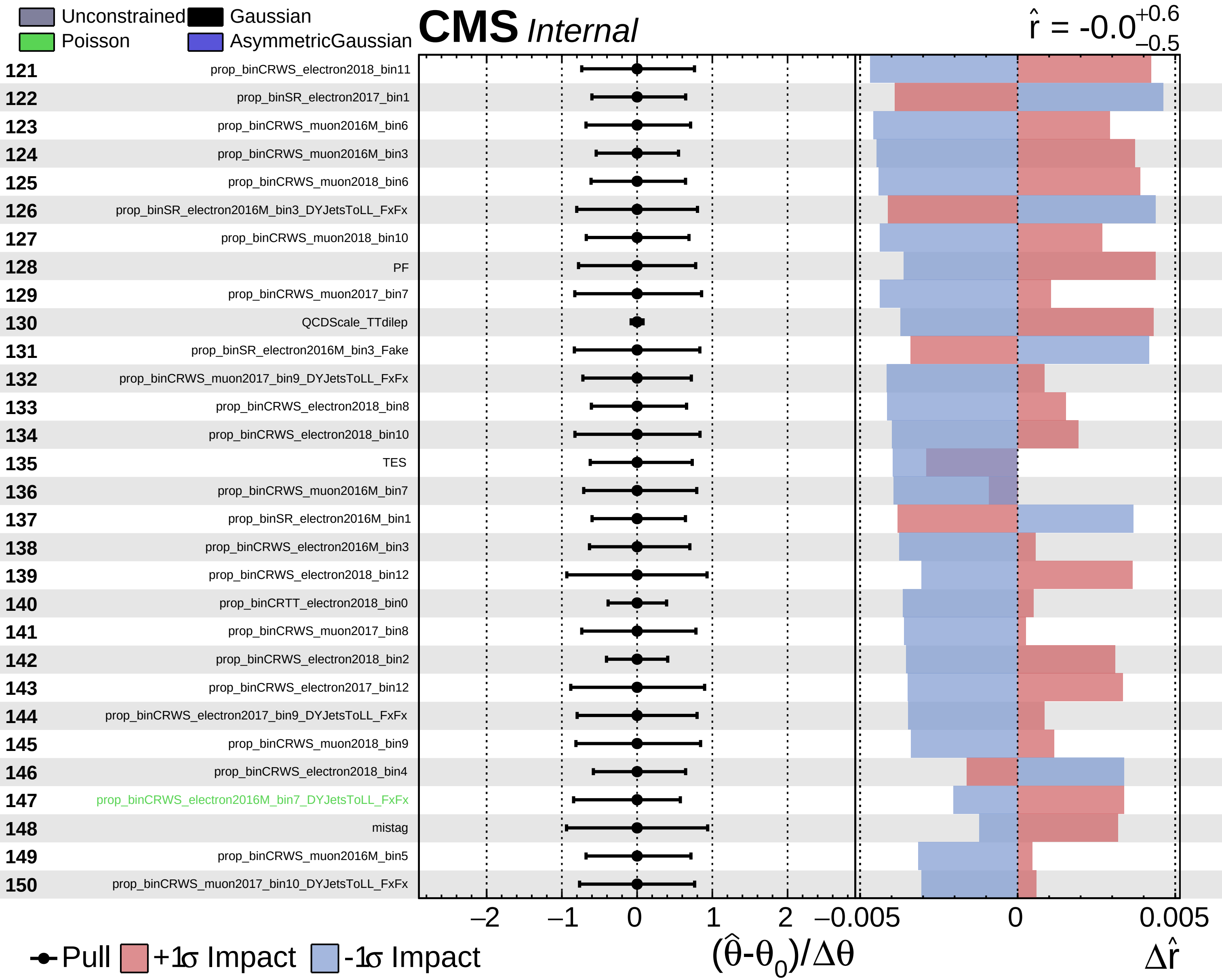
CMS Internal
$\hat{r} = -0.0^{+0.6}_{-0.5}$
1 prop_binSR_muon2018_bin12
2 prop_binSR_electron2018_bin12
3 prop_binSR_muon2017_bin12
4 prop_binSR_muon2018_bin11
5 prop_binSR_muon2016M_bin12
6 prop_binSR_electron2017_bin12
7 prop_binSR_electron2016M_bin12
8 prop_binSR_muon2018_bin10
9 prop_binSR_muon2017_bin11
10 prop_binSR_electron2018_bin11
11 prop_binSR_muon2016M_bin11
12 prop_binSR_muon2018_bin9
13 prop_binSR_muon2017_bin10
14 prop_binSR_muon2018_bin6
15 prop_binSR_electron2018_bin10
16 prop_binSR_electron2017_bin11
17 prop_binSR_muon2018_bin8
18 prop_binSR_muon2016M_bin10
19 prop_binSR_muon2017_bin6
20 prop_binSR_muon2018_bin7
21 prop_binSR_muon2017_bin9
22 prop_binSR_muon2018_bin5
23 prop_binSR_electron2017_bin10
24 prop_binSR_electron2018_bin9
25 prop_binSR_electron2018_bin6
26 prop_binSR_muon2017_bin8
27 prop_binSR_electron2016M_bin11
28 prop_binSR_muon2016M_bin6
29 prop_binSR_muon2016M_bin9
30 prop_binSR_electron2018_bin8
−2 −1 0 1 2
−0.1 0 0.1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

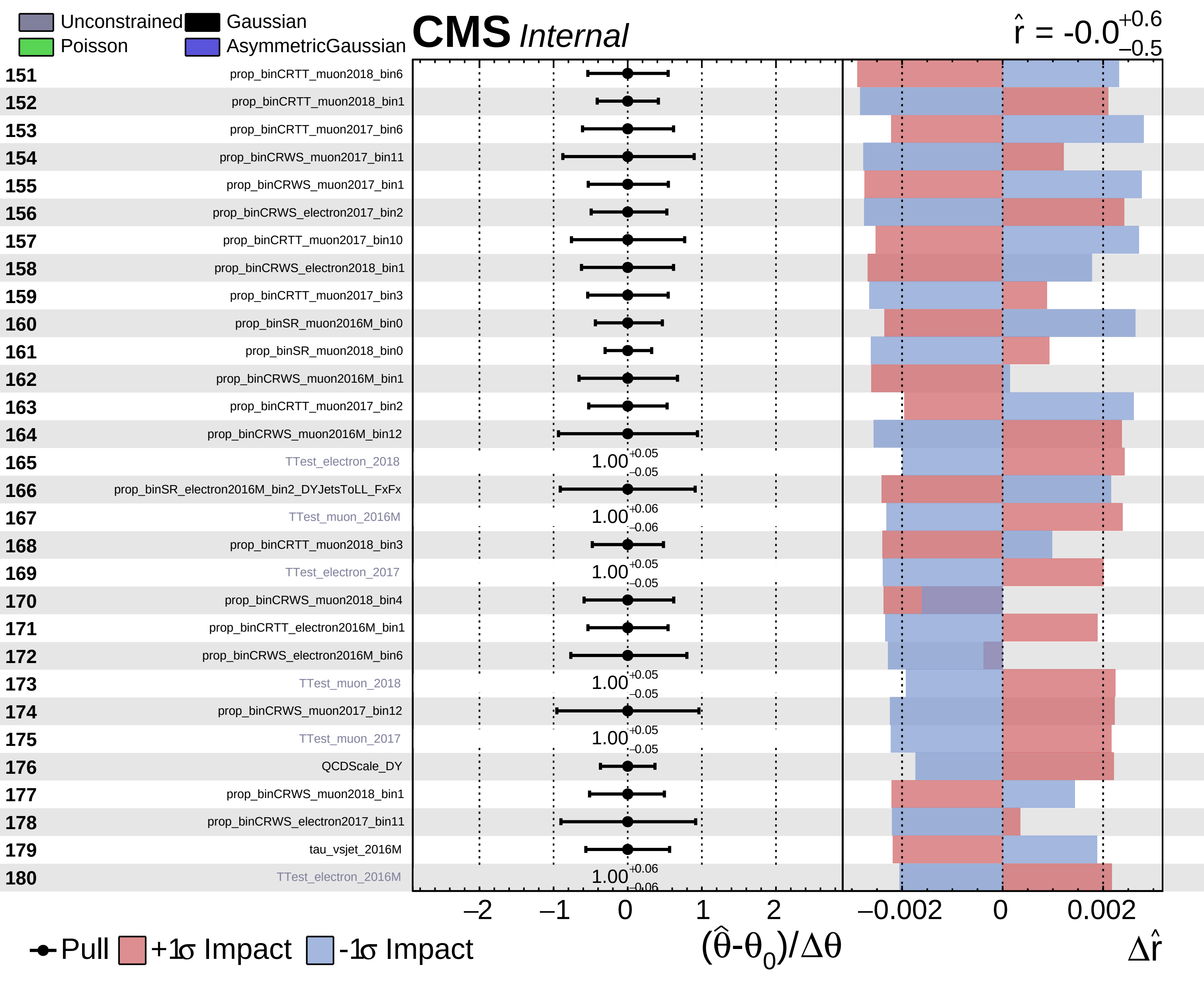

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.0^{+0.6}_{-0.5}$
31 lumi_2018
32 prop_binSR_electron2016M_bin10
33 prop_binSR_electron2017_bin6
34 prop_binSR_muon2017_bin7
35 prop_binSR_muon2017_bin5
36 prop_binSR_electron2017_bin9
37 prop_binSR_electron2018_bin7
38 prop_binSR_muon2016M_bin8
39 prop_binSR_muon2017_bin4
40 prop_binSR_muon2018_bin2
41 prop_binSR_muon2018_bin4
42 prop_binSR_electron2018_bin5
43 prop_binSR_muon2016M_bin5
44 prop_binSR_electron2017_bin8
45 prop_binSR_electron2016M_bin9
46 prop_binSR_muon2016M_bin7
47 QCDScale_WpWpJJ_QCD
48 lumi_2017
49 prop_binSR_muon2018_bin3
50 DYest_muon_2016M $1.00^{+0.34}_{-0.34}$
51 DYest_electron_2018 $1.0^{+0.4}_{-0.4}$
52 prop_binSR_electron2017_bin5
53 DYest_muon_2017 $1.00^{+0.28}_{-0.28}$
54 prop_binSR_electron2018_bin4
55 prop_binSR_muon2017_bin3
56 prop_binSR_electron2016M_bin6
57 prop_binSR_muon2016M_bin4
58 prop_binSR_muon2016M_bin3
59 lumi_2016M
60 DYest_electron_2017 $1.00^{+0.31}_{-0.31}$
−2 −1 0 1 2
−0.02 0 0.02
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.0^{+0.6}_{-0.5}$
61 jer_2018
62 prop_binSR_electron2017_bin7
63 prop_binSR_muon2017_bin2
64 DYest_muon_2018 $1.00^{+0.21}_{-0.21}$
65 prop_binSR_electron2016M_bin8
66 prop_binSR_electron2017_bin4
67 prop_binCRWS_muon2018_bin3
68 prop_binSR_electron2016M_bin5
69 DYest_electron_2016M $1.0^{+0.4}_{-0.4}$
70 prop_binSR_muon2016M_bin2
71 prop_binSR_electron2016M_bin7
72 prop_binCRTT_electron2017_bin0
73 prop_binSR_muon2017_bin1
74 QCDScale_WrongSign
75 prop_binSR_electron2018_bin3
76 prop_binCRWS_electron2018_bin5
77 prop_binCRWS_electron2017_bin4
78 prop_binCRTT_muon2017_bin0
79 prop_binCRWS_muon2016M_bin2
80 prop_binCRWS_electron2018_bin7
81 prop_binSR_electron2016M_bin4
82 prop_binSR_electron2017_bin3
83 prop_binSR_electron2018_bin2
84 prop_binCRTT_electron2018_bin2
85 prop_binSR_muon2018_bin1
86 QCDScale_TVX
87 prop_binCRWS_muon2018_bin2
88 QCDScale_Triboson
89 prop_binCRWS_muon2018_bin8
90 prop_binCRTT_electron2016M_bin0
−2
−1
0
1
2
−0.01
0
0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

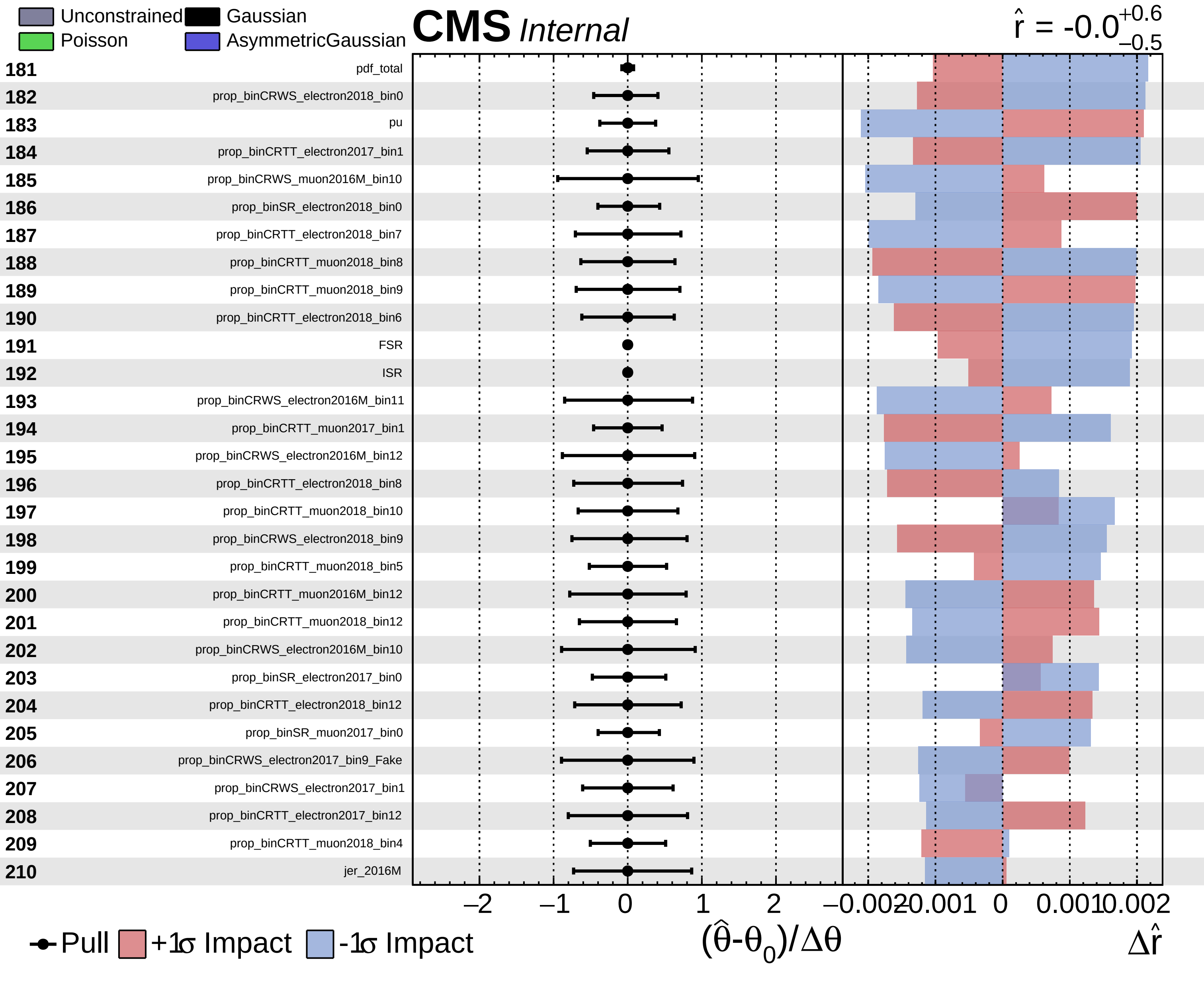


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.0^{+0.6}_{-0.5}$
91 metUnclust
92 prop_binCRTT_muon2016M_bin0
93 prop_binCRWS_electron2018_bin6
94 prop_binCRWS_muon2017_bin3
95 prop_binCRTT_muon2018_bin0
96 prop_binCRWS_muon2018_bin7
97 prop_binCRWS_muon2017_bin5
98 QCDScale_VG
99 prop_binCRWS_electron2016M_bin2
100 prop_binSR_electron2018_bin1
101 prop_binCRWS_muon2016M_bin4
102 prop_binCRWS_muon2018_bin12
103 prop_binCRWS_electron2017_bin8
104 btag
105 prop_binSR_electron2017_bin2
106 jes
107 prop_binSR_muon2016M_bin1
108 prop_binCRWS_muon2017_bin6
109 prop_binCRWS_muon2018_bin0
110 prop_binCRWS_electron2016M_bin0
111 prop_binCRWS_muon2018_bin11
112 prop_binCRWS_muon2016M_bin8_DYJetsToLL_FxFx
113 prop_binCRWS_muon2017_bin0
114 prop_binCRWS_muon2017_bin2
115 prop_binCRTT_muon2016M_bin1
116 prop_binSR_electron2016M_bin2_Fake
117 prop_binCRWS_muon2016M_bin0
118 prop_binCRWS_electron2016M_bin9_DYJetsToLL_FxFx
119 prop_binCRWS_electron2017_bin10
120 prop_binCRWS_muon2017_bin4
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

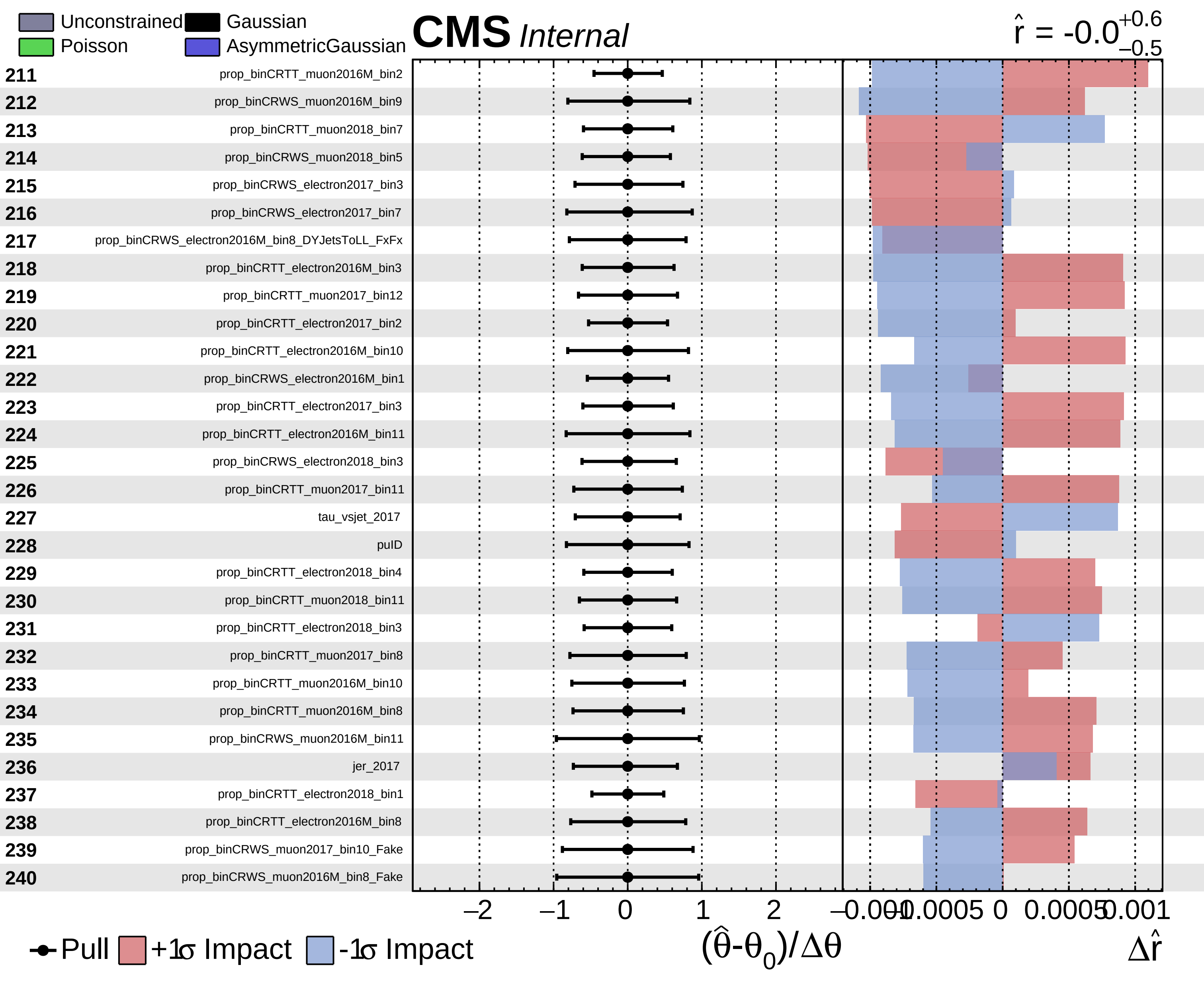

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.0^{+0.6}_{-0.5}$
121 prop_binCRWS_electron2018_bin11
122 prop_binSR_electron2017_bin1
123 prop_binCRWS_muon2016M_bin6
124 prop_binCRWS_muon2016M_bin3
125 prop_binCRWS_muon2018_bin6
126 prop_binSR_electron2016M_bin3_DYJetsToLL_FxFx
127 prop_binCRWS_muon2018_bin10
128 PF
129 prop_binCRWS_muon2017_bin7
130 QCDScale_TTdilep
131 prop_binSR_electron2016M_bin3_Fake
132 prop_binCRWS_muon2017_bin9_DYJetsToLL_FxFx
133 prop_binCRWS_electron2018_bin8
134 prop_binCRWS_electron2018_bin10
135 TES
136 prop_binCRWS_muon2016M_bin7
137 prop_binSR_electron2016M_bin1
138 prop_binCRWS_electron2016M_bin3
139 prop_binCRWS_electron2018_bin12
140 prop_binCRTT_electron2018_bin0
141 prop_binCRWS_muon2017_bin8
142 prop_binCRWS_electron2018_bin2
143 prop_binCRWS_electron2017_bin12
144 prop_binCRWS_electron2017_bin9_DYJetsToLL_FxFx
145 prop_binCRWS_muon2018_bin9
146 prop_binCRWS_electron2018_bin4
147 prop_binCRWS_electron2016M_bin7_DYJetsToLL_FxFx
148 mistag
149 prop_binCRWS_muon2016M_bin5
150 prop_binCRWS_muon2017_bin10_DYJetsToLL_FxFx
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

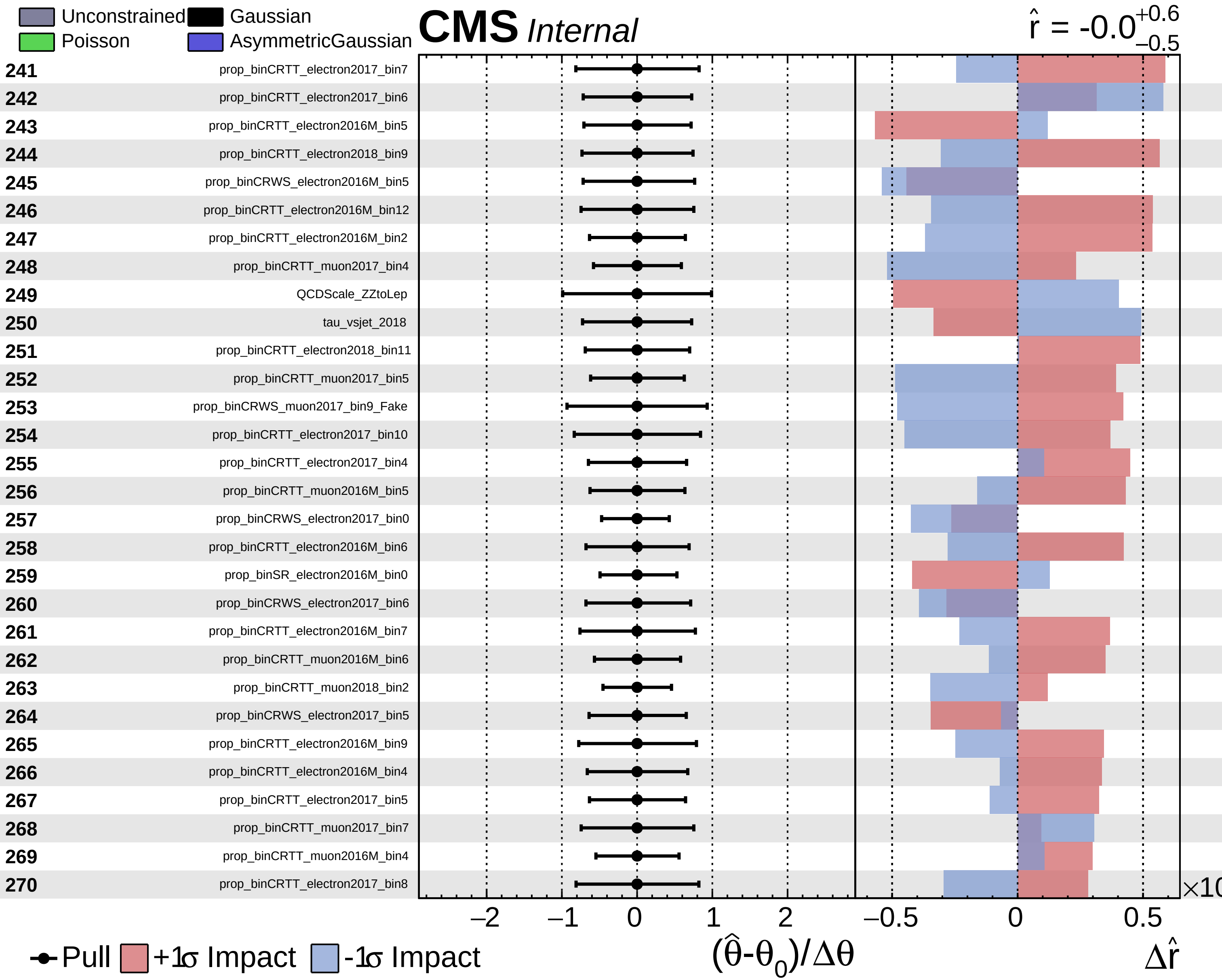


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.0^{+0.6}_{-0.5}$
151 prop_binCRTT_muon2018_bin6
152 prop_binCRTT_muon2018_bin1
153 prop_binCRTT_muon2017_bin6
154 prop_binCRWS_muon2017_bin11
155 prop_binCRWS_muon2017_bin1
156 prop_binCRWS_electron2017_bin2
157 prop_binCRTT_muon2017_bin10
158 prop_binCRWS_electron2018_bin1
159 prop_binCRTT_muon2017_bin3
160 prop_binSR_muon2016M_bin0
161 prop_binSR_muon2018_bin0
162 prop_binCRWS_muon2016M_bin1
163 prop_binCRTT_muon2017_bin2
164 prop_binCRWS_muon2016M_bin12
165 TTest_electron_2018 $1.00^{+0.05}_{-0.05}$
166 prop_binSR_electron2016M_bin2_DYJetsToLL_FxFx
167 TTest_muon_2016M $1.00^{+0.06}_{-0.06}$
168 prop_binCRTT_muon2018_bin3
169 TTest_electron_2017 $1.00^{+0.05}_{-0.05}$
170 prop_binCRWS_muon2018_bin4
171 prop_binCRTT_electron2016M_bin1
172 prop_binCRWS_electron2016M_bin6
173 TTest_muon_2018 $1.00^{+0.05}_{-0.05}$
174 prop_binCRWS_muon2017_bin12
175 TTest_muon_2017 $1.00^{+0.05}_{-0.05}$
176 QCDScale_DY
177 prop_binCRWS_muon2018_bin1
178 prop_binCRWS_electron2017_bin11
179 tau_vsjet_2016M
180 TTest_electron_2016M $1.00^{+0.06}_{-0.06}$
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

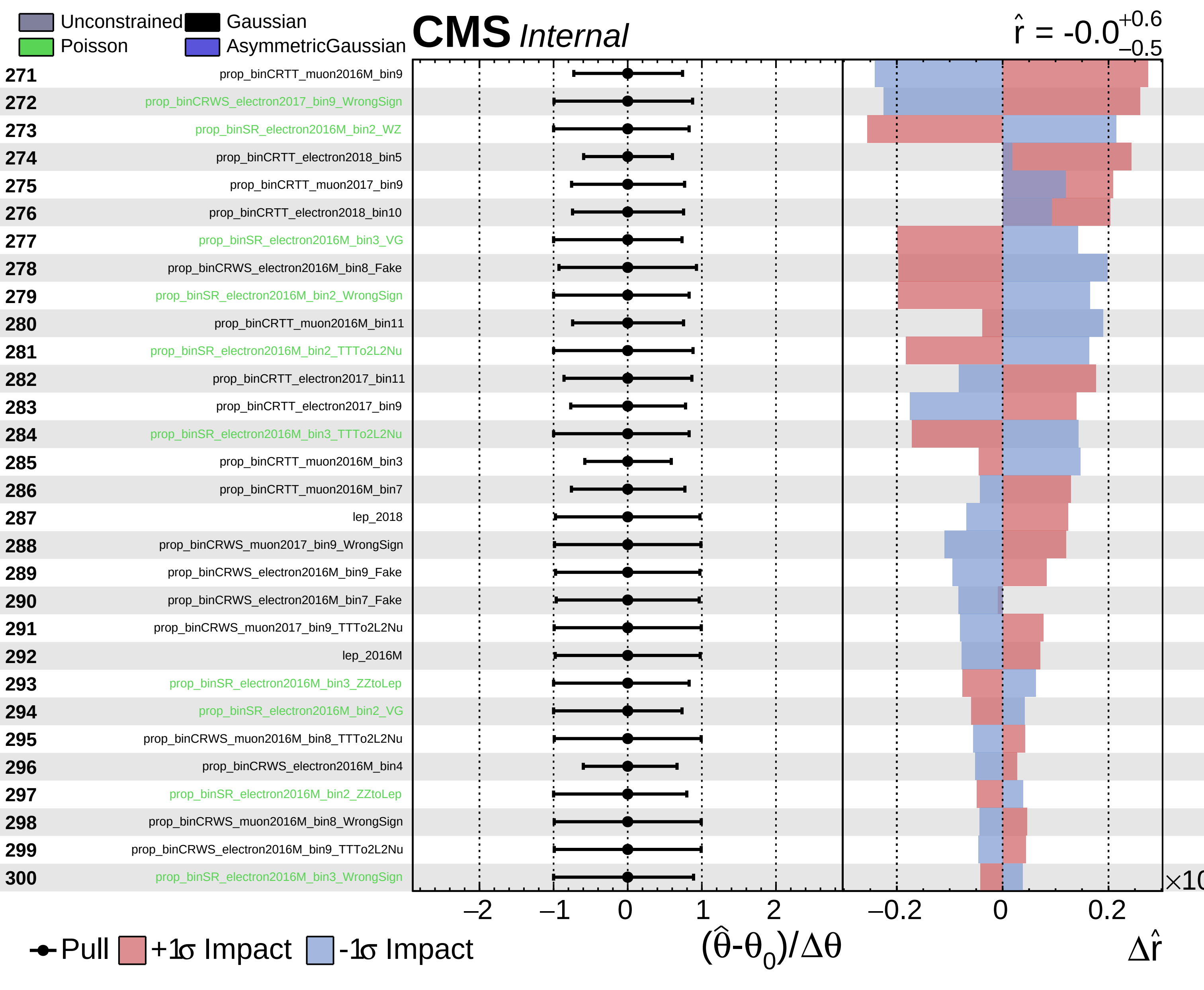


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.0^{+0.6}_{-0.5}$
181 pdf_total
182 prop_binCRWS_electron2018_bin0
183 pu
184 prop_binCRTT_electron2017_bin1
185 prop_binCRWS_muon2016M_bin10
186 prop_binSR_electron2018_bin0
187 prop_binCRTT_electron2018_bin7
188 prop_binCRTT_muon2018_bin8
189 prop_binCRTT_muon2018_bin9
190 prop_binCRTT_electron2018_bin6
191 FSR
192 ISR
193 prop_binCRWS_electron2016M_bin11
194 prop_binCRTT_muon2017_bin1
195 prop_binCRWS_electron2016M_bin12
196 prop_binCRTT_electron2018_bin8
197 prop_binCRTT_muon2018_bin10
198 prop_binCRWS_electron2018_bin9
199 prop_binCRTT_muon2018_bin5
200 prop_binCRTT_muon2016M_bin12
201 prop_binCRTT_muon2018_bin12
202 prop_binCRWS_electron2016M_bin10
203 prop_binSR_electron2017_bin0
204 prop_binCRTT_electron2018_bin12
205 prop_binSR_muon2017_bin0
206 prop_binCRWS_electron2017_bin9_Fake
207 prop_binCRWS_electron2017_bin1
208 prop_binCRTT_electron2017_bin12
209 prop_binCRTT_muon2018_bin4
210 jer_2016M
−2 −1 0 1 2
−0.002 −0.001 0 0.001 0.002
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

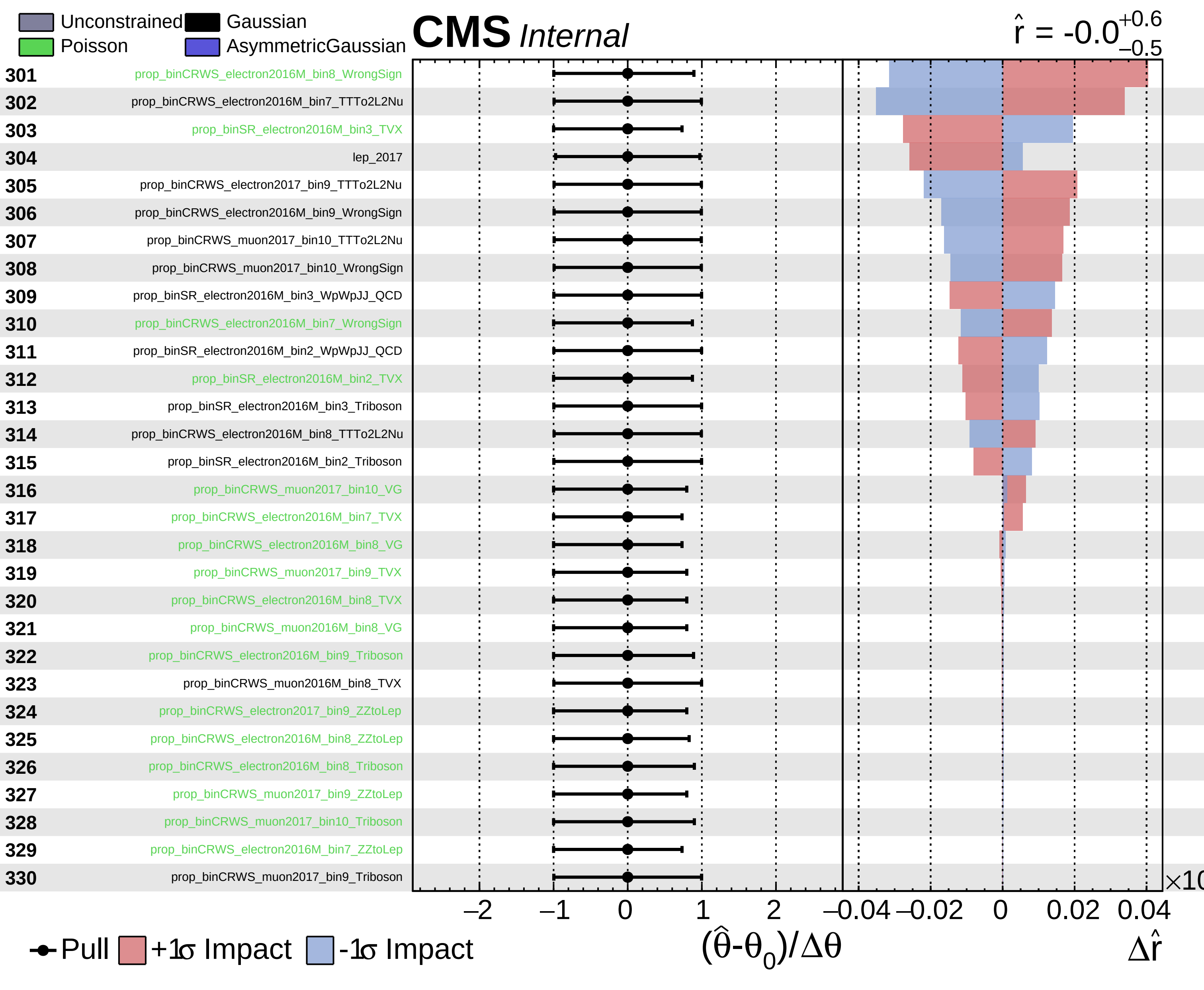

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = -0.0 +0.6 −0.5
211 prop_binCRTT_muon2016M_bin2
212 prop_binCRWS_muon2016M_bin9
213 prop_binCRTT_muon2018_bin7
214 prop_binCRWS_muon2018_bin5
215 prop_binCRWS_electron2017_bin3
216 prop_binCRWS_electron2017_bin7
217 prop_binCRWS_electron2016M_bin8_DYJetsToLL_FxFx
218 prop_binCRTT_electron2016M_bin3
219 prop_binCRTT_muon2017_bin12
220 prop_binCRTT_electron2017_bin2
221 prop_binCRTT_electron2016M_bin10
222 prop_binCRWS_electron2016M_bin1
223 prop_binCRTT_electron2017_bin3
224 prop_binCRTT_electron2016M_bin11
225 prop_binCRWS_electron2018_bin3
226 prop_binCRTT_muon2017_bin11
227 tau_vsjet_2017
228 puID
229 prop_binCRTT_electron2018_bin4
230 prop_binCRTT_muon2018_bin11
231 prop_binCRTT_electron2018_bin3
232 prop_binCRTT_muon2017_bin8
233 prop_binCRTT_muon2016M_bin10
234 prop_binCRTT_muon2016M_bin8
235 prop_binCRWS_muon2016M_bin11
236 jer_2017
237 prop_binCRTT_electron2018_bin1
238 prop_binCRTT_electron2016M_bin8
239 prop_binCRWS_muon2017_bin10_Fake
240 prop_binCRWS_muon2016M_bin8_Fake
−2 −1 0 1 2
−0.001 −0.0005 0 0.0005 0.001
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



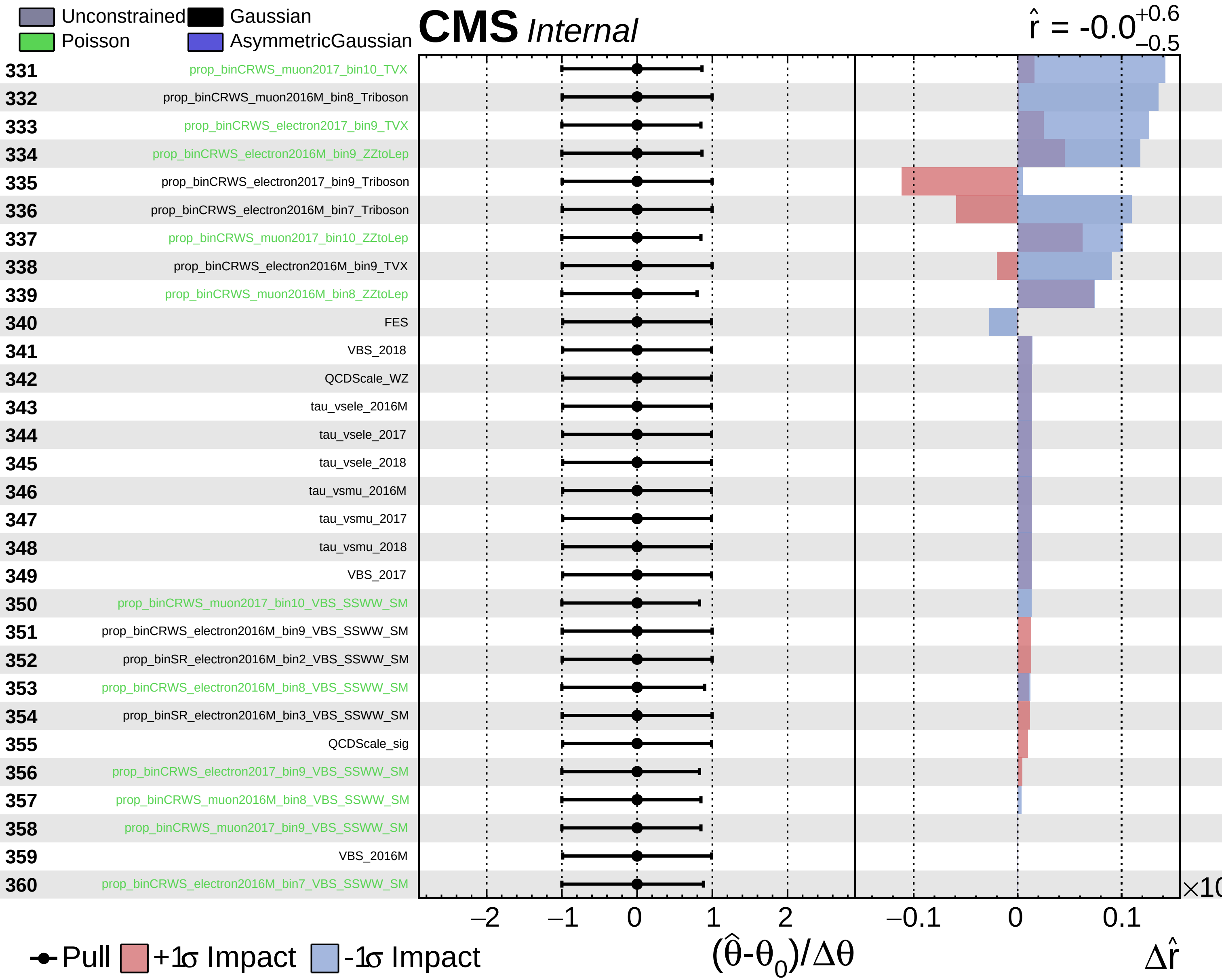
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.0^{+0.6}_{-0.5}$
241 prop_binCRTT_electron2017_bin7
242 prop_binCRTT_electron2017_bin6
243 prop_binCRTT_electron2016M_bin5
244 prop_binCRTT_electron2018_bin9
245 prop_binCRWS_electron2016M_bin5
246 prop_binCRTT_electron2016M_bin12
247 prop_binCRTT_electron2016M_bin2
248 prop_binCRTT_muon2017_bin4
249 QCDScale_ZZtoLep
250 tau_vsjet_2018
251 prop_binCRTT_electron2018_bin11
252 prop_binCRTT_muon2017_bin5
253 prop_binCRWS_muon2017_bin9_Fake
254 prop_binCRTT_electron2017_bin10
255 prop_binCRTT_electron2017_bin4
256 prop_binCRTT_muon2016M_bin5
257 prop_binCRWS_electron2017_bin0
258 prop_binCRTT_electron2016M_bin6
259 prop_binSR_electron2016M_bin0
260 prop_binCRWS_electron2017_bin6
261 prop_binCRTT_electron2016M_bin7
262 prop_binCRTT_muon2016M_bin6
263 prop_binCRTT_muon2018_bin2
264 prop_binCRWS_electron2017_bin5
265 prop_binCRTT_electron2016M_bin9
266 prop_binCRTT_electron2016M_bin4
267 prop_binCRTT_electron2017_bin5
268 prop_binCRTT_muon2017_bin7
269 prop_binCRTT_muon2016M_bin4
270 prop_binCRTT_electron2017_bin8
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.0^{+0.6}_{-0.5}$
271 prop_binCRTT_muon2016M_bin9
272 prop_binCRWS_electron2017_bin9_WrongSign
273 prop_binSR_electron2016M_bin2_WZ
274 prop_binCRTT_electron2018_bin5
275 prop_binCRTT_muon2017_bin9
276 prop_binCRTT_electron2018_bin10
277 prop_binSR_electron2016M_bin3_VG
278 prop_binCRWS_electron2016M_bin8_Fake
279 prop_binSR_electron2016M_bin2_WrongSign
280 prop_binCRTT_muon2016M_bin11
281 prop_binSR_electron2016M_bin2_TTTo2L2Nu
282 prop_binCRTT_electron2017_bin11
283 prop_binCRTT_electron2017_bin9
284 prop_binSR_electron2016M_bin3_TTTo2L2Nu
285 prop_binCRTT_muon2016M_bin3
286 prop_binCRTT_muon2016M_bin7
287 lep_2018
288 prop_binCRWS_muon2017_bin9_WrongSign
289 prop_binCRWS_electron2016M_bin9_Fake
290 prop_binCRWS_electron2016M_bin7_Fake
291 prop_binCRWS_muon2017_bin9_TTTo2L2Nu
292 lep_2016M
293 prop_binSR_electron2016M_bin3_ZZtoLep
294 prop_binSR_electron2016M_bin2_VG
295 prop_binCRWS_muon2016M_bin8_TTTo2L2Nu
296 prop_binCRWS_electron2016M_bin4
297 prop_binSR_electron2016M_bin2_ZZtoLep
298 prop_binCRWS_muon2016M_bin8_WrongSign
299 prop_binCRWS_electron2016M_bin9_TTTo2L2Nu
300 prop_binSR_electron2016M_bin3_WrongSign
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.0^{+0.6}_{-0.5}$
301 prop_binCRWS_electron2016M_bin8_WrongSign
302 prop_binCRWS_electron2016M_bin7_TTTo2L2Nu
303 prop_binSR_electron2016M_bin3_TVX
304 lep_2017
305 prop_binCRWS_electron2017_bin9_TTTo2L2Nu
306 prop_binCRWS_electron2016M_bin9_WrongSign
307 prop_binCRWS_muon2017_bin10_TTTo2L2Nu
308 prop_binCRWS_muon2017_bin10_WrongSign
309 prop_binSR_electron2016M_bin3_WpWpJJ_QCD
310 prop_binCRWS_electron2016M_bin7_WrongSign
311 prop_binSR_electron2016M_bin2_WpWpJJ_QCD
312 prop_binSR_electron2016M_bin2_TVX
313 prop_binSR_electron2016M_bin3_Triboson
314 prop_binCRWS_electron2016M_bin8_TTTo2L2Nu
315 prop_binSR_electron2016M_bin2_Triboson
316 prop_binCRWS_muon2017_bin10_VG
317 prop_binCRWS_electron2016M_bin7_TVX
318 prop_binCRWS_electron2016M_bin8_VG
319 prop_binCRWS_muon2017_bin9_TVX
320 prop_binCRWS_electron2016M_bin8_TVX
321 prop_binCRWS_muon2016M_bin8_VG
322 prop_binCRWS_electron2016M_bin9_Triboson
323 prop_binCRWS_muon2016M_bin8_TVX
324 prop_binCRWS_electron2017_bin9_ZZtoLep
325 prop_binCRWS_electron2016M_bin8_ZZtoLep
326 prop_binCRWS_electron2016M_bin8_Triboson
327 prop_binCRWS_muon2017_bin9_ZZtoLep
328 prop_binCRWS_muon2017_bin10_Triboson
329 prop_binCRWS_electron2016M_bin7_ZZtoLep
330 prop_binCRWS_muon2017_bin9_Triboson
−2 −1 0 1 2
−0.04 −0.02 0 0.02 0.04
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = -0.0 +0.6 −0.5
331 prop_binCRWS_muon2017_bin10_TVX
332 prop_binCRWS_muon2016M_bin8_Triboson
333 prop_binCRWS_electron2017_bin9_TVX
334 prop_binCRWS_electron2016M_bin9_ZZtoLep
335 prop_binCRWS_electron2017_bin9_Triboson
336 prop_binCRWS_electron2016M_bin7_Triboson
337 prop_binCRWS_muon2017_bin10_ZZtoLep
338 prop_binCRWS_electron2016M_bin9_TVX
339 prop_binCRWS_muon2016M_bin8_ZZtoLep
340 FES
341 VBS_2018
342 QCDScale_WZ
343 tau_vsele_2016M
344 tau_vsele_2017
345 tau_vsele_2018
346 tau_vsmu_2016M
347 tau_vsmu_2017
348 tau_vsmu_2018
349 VBS_2017
350 prop_binCRWS_muon2017_bin10_VBS_SSWW_SM
351 prop_binCRWS_electron2016M_bin9_VBS_SSWW_SM
352 prop_binSR_electron2016M_bin2_VBS_SSWW_SM
353 prop_binCRWS_electron2016M_bin8_VBS_SSWW_SM
354 prop_binSR_electron2016M_bin3_VBS_SSWW_SM
355 QCDScale_sig
356 prop_binCRWS_electron2017_bin9_VBS_SSWW_SM
357 prop_binCRWS_muon2016M_bin8_VBS_SSWW_SM
358 prop_binCRWS_muon2017_bin9_VBS_SSWW_SM
359 VBS_2016M
360 prop_binCRWS_electron2016M_bin7_VBS_SSWW_SM
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂